

Variables and Data types in Java

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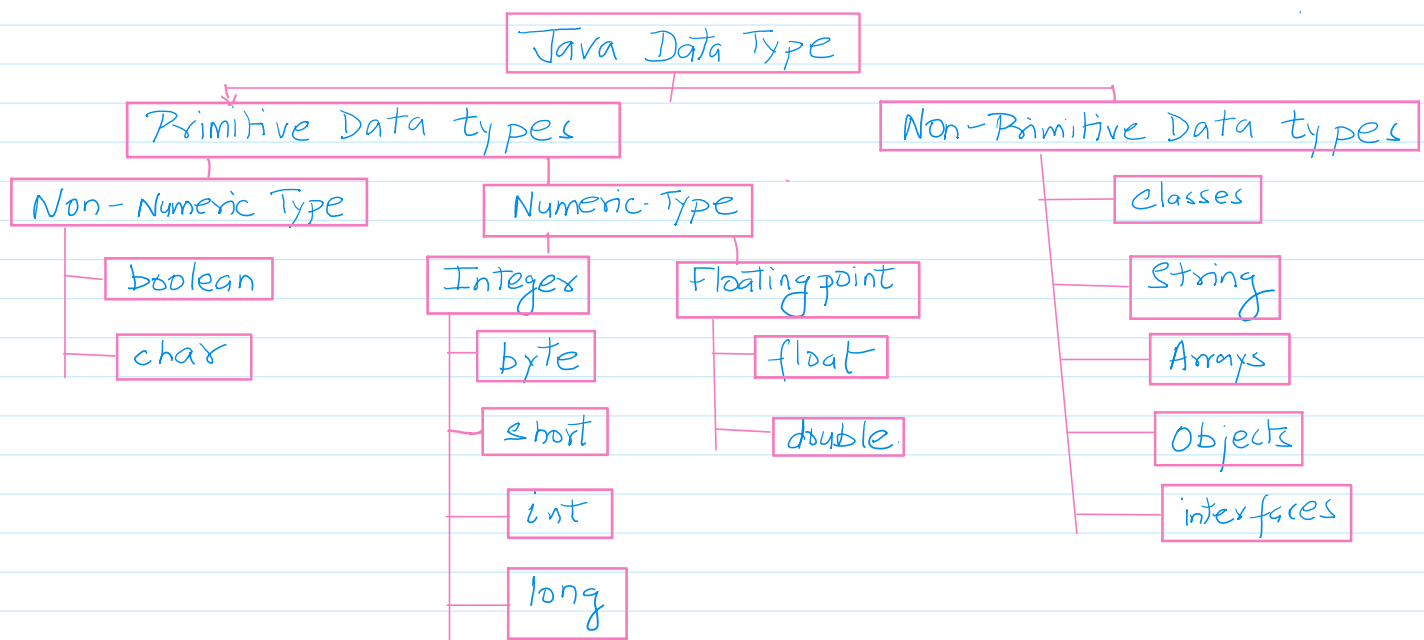
In Java, a variable is the name of a memory location that are used like a container to store values during Java program execution. Every variable must be assigned a data type to define the kind of data it can hold. In Java, variables are categorized into three types: **local**, **instance** and **static** variables.

Syntax to define a Variable in Java:

```
Data_type variable_name;
```

Or

```
Data_type variable_name = value;
```



In Java, primitive data types are the building blocks of data manipulation. These are the basic data types. In Java there are mainly eight primitive data types which are as follows:

- 1) boolean
- 2) char
- 3) byte
- 4) short
- 5) int
- 6) long
- 7) float
- 8) Double

Java is a statically typed programming language. It means all the variables must be declared before their use. That is why we need to declare the variables type and name.

1. **boolean data type:** In Java, the boolean data type represents a single bit of information with 2 possible states: true or false. The size of the boolean data type is one byte (8 bits). It is used to store the result of logical expressions or conditions. Unlike other primitive data types like int or double, boolean does not

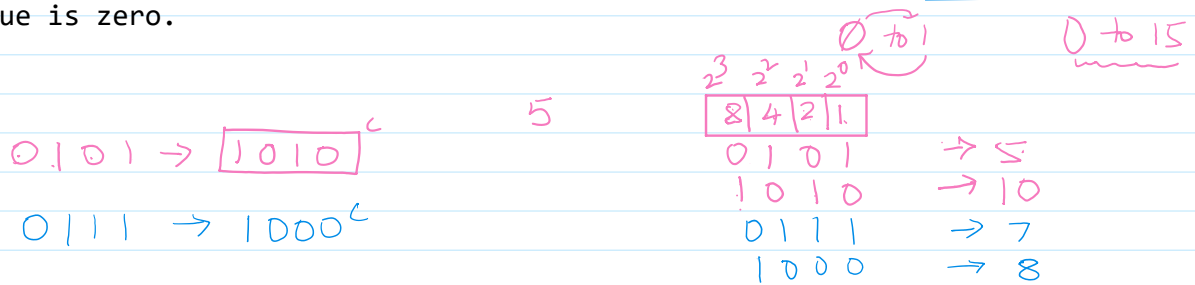
have a specific size or range. It is typically implemented as a single bit, although the exact information may vary across platforms.

For example:

```
boolean isNum;
isNum
false    (boolean) //default value of boolean type

isNum = true; //assignment
isNum
true     (boolean)
```

2. **byte data type:** The byte data type in Java is a primitive data type that represents an 8 bit signed two's complement integer. It has a range of values from -128 to 127. Its default value is zero.



Comfortable room?

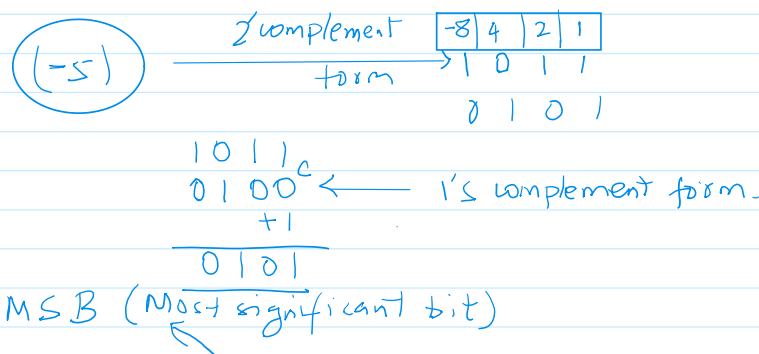
⇒ light
⇒ circulation of Air

APP
location ✓ ✓
location ✓ ✓

L	F
0	0
1	0
0	1
1	1

IOT
Internet of Thing

Two's complement is used to represent (-ve) numbers.



For example:

byte b;

b

0 (byte)

b = -5;

b

-128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1

→ -128
→ 127

256 types of values we get.

```

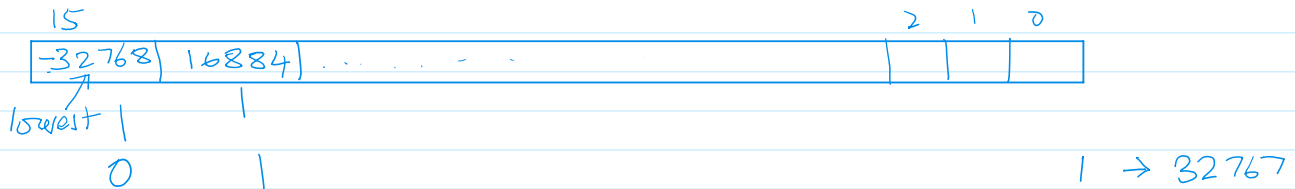
byte b = -5;
b = 128;

```

-5 (byte) → need more bits, then the byte data type allocates. We can declare the variable type as short or int.

Error: incompatible types: possible lossy conversion from int to byte

3. **short Data type:** The short data type in Java is a primitive data type that represents a 16-bit signed two's complement integer. Its range of value is -32,768 to 32,767. Similar to the byte data type, short is used when memory conservation is a concern but more precision than byte is required. Its default value is zero.



For example:

```

short s;
s = 128;
s = 32768;

```

128 (short) → not stored in s variable; it needs higher data type.

Error: incompatible types: possible lossy conversion from int to short

s = 128 (short) → old value.

4. **int Data type:** The int data type in Java is a primitive data type that represents a 32 bit signed two's complement integer. It occupies 4 byte of data. It has a range of values from -2147483648 to 2147483647. The int data type is one of the most commonly used data types. It is typically used to store whole numbers without decimal points. Its default value is zero.

```

int i = 32768;
int j;
j = 2147483648;

```

32768 (int) → need more space to hold this value.

Error: integer number too large

5. **long Data Type:** The long data type in Java is a primitive data type that represents a 64 - bits signed two's complement integer. It has a wider range of values than int, ranging from $(-1) \times 2^{63}$ to $(2^{63}) - 1$.

-9223372036854775808 to 9223372036854775807.

Its default value is 0L or 0l.

For example:

```
long l;
```

```
l
```

```
0 (long)
```

```
l = 2147483648;
```

```
Error: integer number too large
```

```
l = 2147483648L;
```

```
2147483648 (long)
```

```
l = 9223372036854775807l
```

```
9223372036854775807 (long)
```

```
l = 9223372036854775808l
```

```
Error: integer number too large
```

→ we can store this value approximately in float or double data type.

The long data type is commonly used in application where large integer values are required, such as in scientific computation, financial applications, and system programming. It provides greater precision and a larger range than int, making it suitable for scenarios where int is insufficient.

6. **float Data type:** The float data type in Java is a primitive data type that represents single precision 32-bit IEEE754 floating-point numbers. It can represent a wide range of decimal values, but it is not suitable for precise values such as currency. Its default value is 0.0f or 0.0F.

The float data type is useful for applications where a higher range of values is needed and precision is not critical. It supports 6 decimal places approximation.

For example:

```
float f;
```

```
f
```

```
0.0 (float)
```

```
f = 9223372036854775808f;
```

```
f
```

```
9.223372E18 (float)
```

7. **double Data type:** The double data type in Java is a primitive data type that represents double - precision, 64 - bits IEEE 754 floating - point numbers. Its default value is 0.0. It provides a wider range of values and greater precision compared to the float data type. Which makes it suitable for applications where accurate representation of decimal values is required. It supports 15 decimal places approximation. For example:

```
double d;
```

```
d
```

```
0.0 (double)
```

```
d = 9223372036854775808;
```

```
Error: integer number too large
```

```
d = 9223372036854775808D;
```

```
d
```

```
9.223372036854776E18 (double)
```

8. **char Data type:** The character type in Java is a primitive data type that represents a single 16 bit Unicode character. It can store any character from the Unicode character set, which allows Java to support the internationalization and representation of characters from various languages and writing systems.

```

char c;
c
'\u0000'    (char)
c = 'd';
c
'd'    (char)
c = '\u20b9';
c
'₹'    (char)

```

Type	Default values	Size	Range of Values
boolean	false	1 bit	true, false
byte	0	8 bits	-128 to 127
char	'\u0000'	16-bits	Any Unicode character
short	0	16-bits	-32768 to 32767
int	0	32-bits	-2147483648 to 2147483647
long	0	64-bits	-9223372036854775808 to 9223372036854775807
float	0.0f or 0.0F	32-bits	Up to 6 decimal places
double	0.0	64-bits	Up to 15 decimal places

String data type: It is basically a class type, use to hold multiple characters inside double quotes. It is not a primitive type. It is a class type.

For example:

```
String str;
```

```
str
```

```
null
```

```
str = "Jiya";
```

```
str
```

```
"Jiya"    (String)
```

→ default value of String type.

```

/**
 * Write a description of class PrimitiveDataTypes here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class PrimitiveDataTypes
{
    public static void main(String[] args){
        boolean b = true;
        byte bt = 10;
        char c = '\u20b9';
        short s = 3456;
        int i = 3545678;
        float f = 3.14f;
        double d = 0.987654323456;
        long l = 1234567897654345678L;
        String str = "Hello!  Java";

        System.out.print("\fb = " + b +

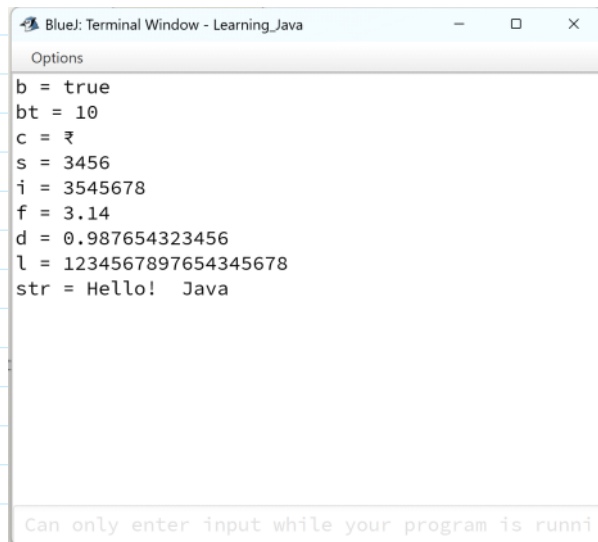
```

```

        "\nbt = " + bt +
        "\nc = " + c +
        "\ns = " + s +
        "\ni = " + i +
        "\nf = " + f +
        "\nd = " + d +
        "\nl = " + l +
        "\nstr = " + str);
    }
}

```

Output:



```

BlueJ: Terminal Window - Learning_Java
Options
b = true
bt = 10
c = ₹
s = 3456
i = 3545678
f = 3.14
d = 0.987654323456
l = 1234567897654345678
str = Hello! Java
Can only enter input while your program is running

```

Home work

Fill in the Blanks (10 Questions)

1. In Java, variables are containers used to store _____ values.
2. Java variables are categorized into _____ types.
3. The default value of boolean data type is _____.
4. The _____ data type is used to store a single character.
5. The size of int data type in Java is _____ bytes.
6. The float data type should end with suffix _____.
7. The double data type is used for storing _____ point numbers.
8. The byte data type has a range from _____ to 127.
9. Strings in Java are enclosed within _____ quotes.
10. The long data type should end with suffix _____.

True / False Questions (10 Questions)

1. Java supports both primitive and non-primitive data types.
2. The char data type stores multiple characters.
3. The size of short data type is 2 bytes.
4. float provides more precision than double.
5. The default value of numeric primitive data types is 0.
6. A boolean variable can store only true or false values.
7. The byte data type occupies 8 bits in memory.
8. Strings are primitive data types in Java.
9. The long data type can store larger values than int.

10. Java is a strongly typed programming language.

Multiple Choice Questions (MCQ) (10 Questions)

1. Which of the following is NOT a primitive data type in Java?

- A. int
- B. float
- C. String
- D. char

2. What is the size of byte data type?

- A. 1 bit
- B. 1 byte
- C. 2 bytes
- D. 4 bytes

3. Which keyword is used to declare a constant value in Java?

- A. static
- B. final
- C. const
- D. fixed

4. Which data type is used to store decimal numbers with high precision?

- A. int
- B. float
- C. double
- D. byte

5. Which of the following can store only true or false values?

- A. int
- B. char
- C. boolean
- D. byte

6. What is the default value of a boolean variable?

- A. true
- B. false
- C. 0
- D. null

7. Which symbol is used to represent string literals in Java?

- A. ' '
- B. < >
- C. " "
- D. ()

8. Which data type is suitable for storing age?

- A. boolean
- B. int
- C. char
- D. float

9. Which of the following data types occupies 16 bits?

- A. byte
- B. short
- C. int

D. long

10. Which of the following is a non-primitive data type?

A. float

B. char

C. String

D. boolean